

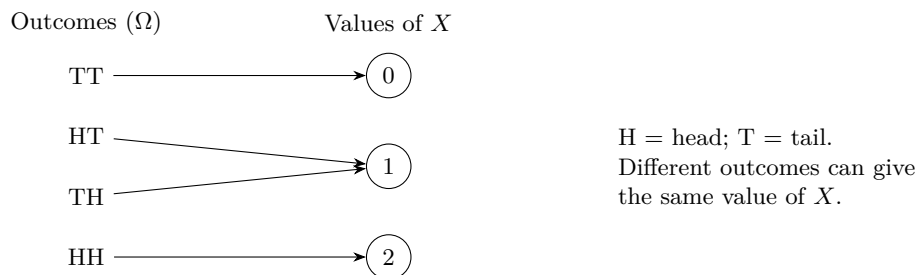
1 Start here: what is a random variable?

Scope: random variables; discrete and continuous random variables (DRV/CRV); expectation, mean, variance and properties of expectation; binomial and Poisson distributions; the one-sample t -test. These notes contain **13 fully worked questions** and reproducible black-and-white figures.

A **random experiment** has an uncertain outcome. Its **sample space** Ω is the set of all possible outcomes. An **event** is a set of outcomes. A **random variable** assigns one real number to each outcome:

$$X : \Omega \longrightarrow \mathbb{R}.$$

Here Ω is the sample space, $\mathbb{R}$ is the set of real numbers, X is the rule, and lowercase x denotes a possible numerical value of X . For example, toss two coins and let X count heads:



Discrete versus continuous: the essential distinction

	DRV	CRV (with a density)
Possible values	Finite or countably infinite, e.g. $0, 1, 2, \dots$	Values over a continuum, e.g. time in an interval
Description	PMF: $p_X(x) = P(X = x)$	PDF: $f_X(x)$, a probability <i>density</i>
Validity	$p_X(x) \geq 0$, $\sum_x p_X(x) = 1$	$f_X(x) \geq 0$, $\int_{-\infty}^{\infty} f_X(x) dx = 1$
Probability	Add masses at allowed values	Integrate density over an interval
Single point	$P(X = x)$ may be positive	$P(X = x) = 0$
Graph	Separate stems/bars at values	Curve; probability is area underneath

Here P means probability, $\sum_x$ means “sum over all possible values of X ,” and $\int f_X(x) dx$ means area under the density. The subscript X identifies the random variable; it is often omitted when there is no ambiguity. The **support** is the set/range of possible values with positive mass/density (including boundary points when convenient).

The cumulative distribution function (CDF): works for both

$$\boxed{F_X(x) = P(X \leq x)} \quad F_X(x) = \begin{cases} \sum p_X(u) & \text{DRV,} \\ \int_{-\infty}^x f_X(u) du & \text{CRV.} \end{cases}$$

Here x is the cutoff, u is a dummy summation/integration variable, and F_X is cumulative probability. Every CDF is nondecreasing, right-continuous, lies between 0 and 1, tends to 0 on the far left and to 1 on the far right. For any $a < b$,

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

For a CRV, endpoint inclusion does not change probability. For a DRV it can: use the actual included values.

Exam workflow: define X and its support $\rightarrow$ state the model and assumptions $\rightarrow$ write the formula and define its symbols $\rightarrow$ substitute $\rightarrow$ calculate $\rightarrow$ interpret. For a graph: label both axes, mark values/cutoffs, and shade or highlight the requested probability.

2 DRV: probability, expectation, mean and variance

The **expectation** or **mean** is the probability-weighted average. The **variance** measures average squared distance from the mean; the **standard deviation** is its square root.

$$\mu = E[X] = \sum_x x p_X(x), \quad E[X^2] = \sum_x x^2 p_X(x), \quad \sigma^2 = \text{Var}(X) = E[X^2] - \mu^2, \quad \sigma = \sqrt{\sigma^2}.$$

Here μ is the distribution mean, σ^2 its variance, σ its standard deviation, x an allowed value, and $p_X(x)$ its probability. In general,

$$E[g(X)] = \sum_x g(x) p_X(x), \quad \text{Var}(X) = \sum_x (x - \mu)^2 p_X(x),$$

where g is a specified function. These formulas require the relevant sums to exist. Mean and SD have the same units as X ; variance has squared units.

Worked question 1: Find the constant, probabilities, mean, variance and CDF

Question. $P(X = x) = k(x + 1)$ for $x = 0, 1, 2$ and zero otherwise. Find k , $P(X \geq 1)$, $E[X]$, $\text{Var}(X)$, $\text{SD}(X)$ and $F_X(x)$.

Formula first. $\sum_x p_X(x) = 1$; $P(X \geq a) = \sum_{x \geq a} p_X(x)$; $\mu = \sum_x x p_X(x)$; $\sigma^2 = \sum_x x^2 p_X(x) - \mu^2$; $\sigma = \sqrt{\sigma^2}$; $F_X(x) = \sum_{u \leq x} p_X(u)$. **Symbols.** k is the unknown normalizing constant; $x, u \in \{0, 1, 2\}$ are values of X ; $a = 1$ is the cutoff; p_X is the PMF; μ, σ^2, σ are mean, variance and SD.

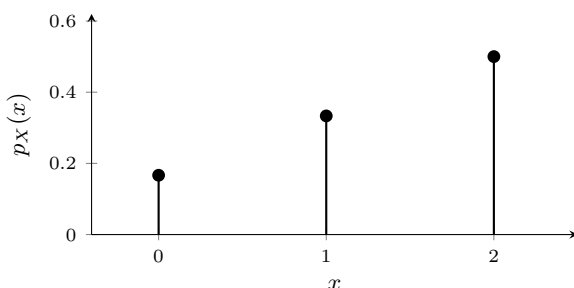
$$k(1 + 2 + 3) = 1 \Rightarrow k = \frac{1}{6}, \quad P(X \geq 1) = \frac{2}{6} + \frac{3}{6} = \frac{5}{6}.$$

x	$p_X(x)$	$x p_X(x)$	$x^2 p_X(x)$
0	1/6	0	0
1	2/6	2/6	2/6
2	3/6	6/6	12/6
Sum	1	4/3	7/3

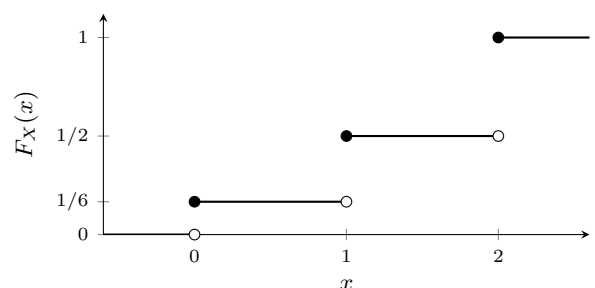
$$\mu = \frac{4}{3}, \quad \sigma^2 = \frac{7}{3} - \left(\frac{4}{3}\right)^2 = \frac{5}{9}, \quad \sigma = \frac{\sqrt{5}}{3} \approx 0.7454.$$

$$F_X(x) = \begin{cases} 0 & x < 0, \\ 1/6 & 0 \leq x < 1, \\ 1/2 & 1 \leq x < 2, \\ 1 & x \geq 2. \end{cases}$$

PMF: masses sum to 1



CDF: jump size = point probability



Graph marks: Use filled dots at included CDF endpoints and open dots at excluded endpoints. A PMF is not a smooth density curve.

The mean need not be a possible observation: here $\mu = 4/3$ although X only takes 0, 1, 2. Never replace $E[X^2]$ by $(E[X])^2$.

3 CRV: density, area, expectation and variance

For a CRV, density height is *not* probability. A density may exceed 1, but its total area must be 1. With density f_X , endpoints $a < b$, and a function g ,

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx, \quad E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

Thus $\mu = \int_{-\infty}^{\infty} x f_X(x) dx$, $E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$ and $\sigma^2 = E[X^2] - \mu^2$, with the integrals over the whole support. Here μ, σ^2 are the mean and variance, and x is the integration variable. For a differentiable CDF, $f_X(x) = F'_X(x)$.

Worked question 2: Normalize a density and calculate all basic quantities

Question. Let $f_X(x) = kx$ for $0 \leq x \leq 1$, and 0 otherwise. Find k , $P(1/4 \leq X \leq 3/4)$, the mean, variance, SD and CDF.

Formula first.

$$\int f_X(x) dx = 1, \quad P(a \leq X \leq b) = \int_a^b f_X(x) dx, \quad \mu = \int x f_X(x) dx, \quad \sigma^2 = \int x^2 f_X(x) dx - \mu^2,$$

$$\sigma = \sqrt{\sigma^2}, \quad F_X(x) = \int_{-\infty}^x f_X(u) du.$$

Symbols. k is the unknown constant; f_X is the density; $a = 1/4$, $b = 3/4$ are interval bounds; μ, σ^2, σ are mean, variance and SD; u is a dummy integration variable. All unlabelled integrals above cover the whole support $[0, 1]$.

$$1 = \int_0^1 kx dx = k[x^2/2]_0^1 = k/2 \Rightarrow \boxed{k = 2},$$

$$P(1/4 \leq X \leq 3/4) = \int_{1/4}^{3/4} 2x dx = [x^2]_{1/4}^{3/4} = 9/16 - 1/16 = \boxed{1/2},$$

$$\mu = \int_0^1 x(2x) dx = 2[x^3/3]_0^1 = \boxed{2/3},$$

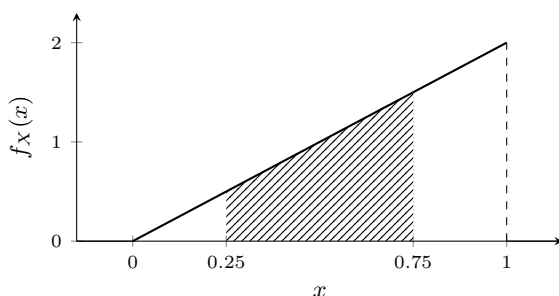
$$E[X^2] = \int_0^1 x^2(2x) dx = 2[x^4/4]_0^1 = 1/2,$$

$$\sigma^2 = 1/2 - (2/3)^2 = \boxed{1/18}, \quad \boxed{\sigma = 1/\sqrt{18} \approx 0.2357}.$$

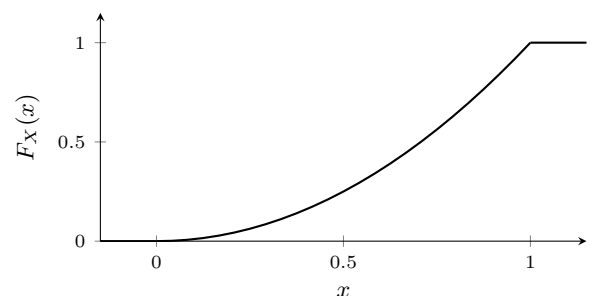
The integration rule used is $\int x^m dx = x^{m+1}/(m+1) + C$ for $m \neq -1$, where m is the power and C the integration constant. Brackets $[h(x)]_a^b$ mean $h(b) - h(a)$.

$$F_X(x) = \begin{cases} 0 & x < 0, \\ \int_0^x 2u du = x^2 & 0 \leq x \leq 1, \\ 1 & x > 1. \end{cases}$$

PDF: hatched area = 1/2



CDF: continuous and nondecreasing



Graph marks: Label the density $2x$, the endpoints 0 and 1, and the requested interval. Hatch the area beneath the curve, not above it. The dashed boundary is a guide, not part of the density.

4 Properties of expectation and variance

Here $|a|$ means the absolute (nonnegative) value of a . Let X, Y be random variables, a, b, c fixed real constants, and $\mu_X = E[X]$, $\mu_Y = E[Y]$. Assume the required expectations are finite.

Formula	Meaning / condition
$E[c] = c$	The average of a constant is itself.
$E[aX + b] = aE[X] + b$	Scale and shift the mean.
$E[aX + bY + c] = aE[X] + bE[Y] + c$	Linearity; independence is <i>not</i> required.
$E[XY] = E[X]E[Y]$ if X, Y independent	Do not apply this to arbitrary dependent variables.
$\text{Var}(X) = E[X^2] - (E[X])^2$	Second moment minus squared mean.
$\text{Var}(c) = 0$; $\text{Var}(aX + b) = a^2 \text{Var}(X)$	A shift leaves spread unchanged.
$\text{SD}(aX + b) = a \text{SD}(X)$	SD cannot be negative.
$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ if independent	Without independence, include covariance.

For completeness, $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$ measures joint variation and $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$. Independent variables have zero covariance.

Why the shortcut for variance works:

$$\text{Var}(X) = E[(X - \mu_X)^2] = E[X^2] - 2\mu_X E[X] + \mu_X^2 = E[X^2] - \mu_X^2.$$

Why linearity works (DRV): $E[aX + b] = \sum_x (ax + b)p_X(x) = a \sum_x xp_X(x) + b \sum_x p_X(x) = aE[X] + b$. For a CRV, replace the sums by integrals and use $\int f_X(x) dx = 1$.

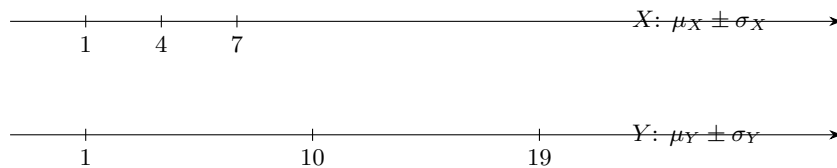
Worked question 3: A linear transformation

Question. X has mean 4 and variance 9. If $Y = 3X - 2$, find $E[Y]$, $\text{Var}(Y)$ and $\text{SD}(Y)$.

Formula first. $E[aX + b] = a\mu_X + b$; $\text{Var}(aX + b) = a^2\sigma_X^2$; $\text{SD}(Y) = \sqrt{\text{Var}(Y)}$. **Symbols.** $a = 3$ is the multiplier, $b = -2$ the shift, $\mu_X = 4$ the mean, $\sigma_X^2 = 9$ the variance, and Y the transformed variable.

$$E[Y] = 3(4) - 2 = \boxed{10}, \quad \text{Var}(Y) = 3^2(9) = \boxed{81}, \quad \text{SD}(Y) = \sqrt{81} = \boxed{9}.$$

Spread is multiplied by 3; subtracting 2 only shifts.



The marked positions show mean and mean $\pm$ one SD, not guaranteed observation limits.

Worked question 4: Expectation of a nonlinear function

Question. $P(X = 0) = 1/4$, $P(X = 2) = 3/4$. Find $E[X]$, $E[X^2]$ and $E[2X^2 + 3X + 1]$.

Formula first. $E[X] = \sum_x xp_X(x)$; $E[X^2] = \sum_x x^2 p_X(x)$; $E[aX^2 + bX + c] = aE[X^2] + bE[X] + c$. **Symbols.** $x \in \{0, 2\}$ is an allowed value; $p_X(x)$ is its probability; $a = 2, b = 3, c = 1$ are fixed coefficients.

$$E[X] = 0(1/4) + 2(3/4) = \frac{3}{2}, \quad E[X^2] = 0^2(1/4) + 2^2(3/4) = 3,$$

$$E[2X^2 + 3X + 1] = 2(3) + 3(3/2) + 1 = \boxed{\frac{23}{2} = 11.5}.$$

Common trap: generally $E[g(X)] \neq g(E[X])$. Here $E[X^2] = 3$ but $(E[X])^2 = 9/4$. Linearity applies to sums and constant multiples; it does not let you move expectation through a square.

5 Binomial distribution: count successes in fixed trials

Write $X \sim \text{Bin}(n, p)$ when X is the number of successes in n trials satisfying **all four** conditions:

1. A fixed number n of trials.
2. Each trial has two categories: success and failure (“success” is the outcome being counted).
3. Trials are independent.
4. Success probability p is the same in every trial; failure probability is $q = 1 - p$.

$$P(X = r) = \binom{n}{r} p^r q^{n-r}, \quad r = 0, 1, \dots, n; \quad \mu = np, \quad \sigma^2 = npq, \quad \sigma = \sqrt{npq}.$$

Here n is a nonnegative integer, r is the requested number of successes, $0 \leq p \leq 1$, $q = 1 - p$, and μ, σ^2, σ are the mean, variance and SD. The combination count is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad n! = n(n-1) \cdots 1, \quad 0! = 1.$$

For a particular ordering, independence gives probability $p^r q^{n-r}$. There are $\binom{n}{r}$ different orderings with r successes; multiplying gives the PMF. Outside $0, \dots, n$, the probability is zero.

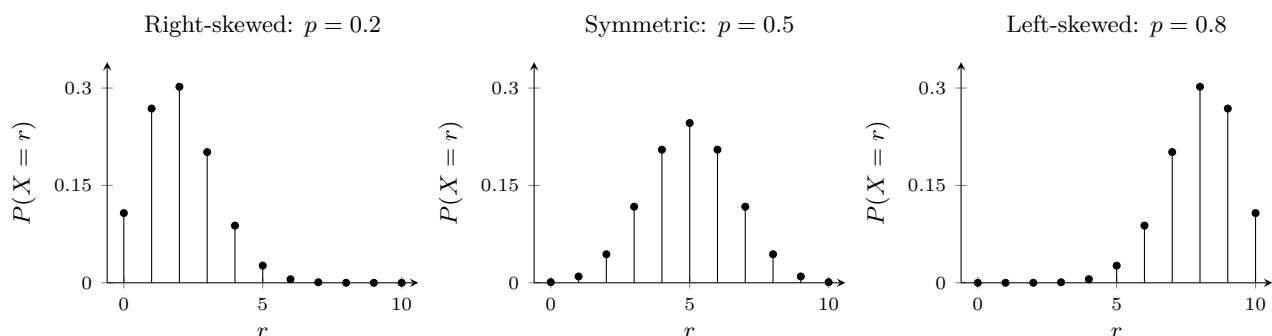
Translate exam wording before calculating

In this table k is an integer cutoff, r a summation index and $X \sim \text{Bin}(n, p)$.

Wording	Formula
Exactly k	$P(X = k) = \binom{n}{k} p^k q^{n-k}$
At most k / no more than k	$P(X \leq k) = \sum_{r=0}^k \binom{n}{r} p^r q^{n-r}$
Fewer than k	$P(X < k) = \sum_{r=0}^{k-1} \binom{n}{r} p^r q^{n-r}$
At least k	$P(X \geq k) = 1 - \sum_{r=0}^{k-1} \binom{n}{r} p^r q^{n-r}$
More than k	$P(X > k) = 1 - \sum_{r=0}^k \binom{n}{r} p^r q^{n-r}$
At least one	$P(X \geq 1) = 1 - P(X = 0) = 1 - q^n$

An empty sum is zero. If a cutoff lies outside the support, use the support first: $X \leq n$ always.

How p changes the shape ($n = 10$ in all three)



Graph marks: Use isolated masses, integer ticks, labelled axes and the parameters n, p . A right-skewed distribution has its longer tail on the right. At $p = 0$ or 1 , all mass is at one endpoint.

Short derivation of the mean and variance

Let $I_i = 1$ for success on trial i and 0 for failure, where $i = 1, \dots, n$. Then $X = \sum_{i=1}^n I_i$, $E[I_i] = p$, and $\text{Var}(I_i) = p - p^2 = pq$. Linearity gives $E[X] = np$; independence gives $\text{Var}(X) = npq$. A single such trial is a **Bernoulli** random variable.

Sampling without replacement from a small finite population generally breaks independence and constant p . Do not assume a binomial model just because the outcome is success/failure.

6 Binomial numericals: direct, inverse and minimum trials

Worked question 5: Exactly, at most and at least

Question. Each of five independent items is defective with probability 0.4. Find the probabilities of exactly two, at most one, and at least one defective, and find the mean and variance of the defective count.

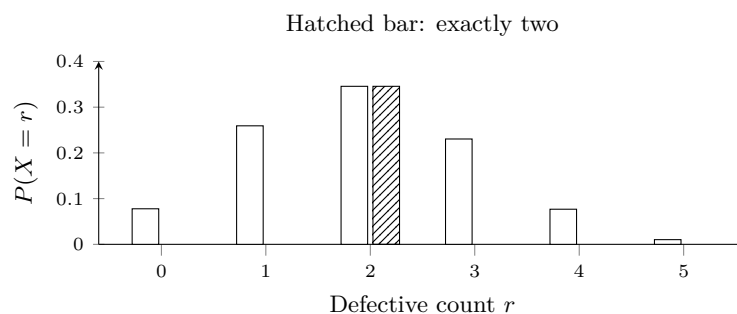
Formula first. $P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$; $P(X \leq 1) = P(X = 0) + P(X = 1)$; $P(X \geq 1) = 1 - (1-p)^n$; $\mu = np$, $\sigma^2 = np(1-p)$. **Symbols.** X counts defectives; $n = 5$ items; $p = 0.4$ is defect probability; $q = 1 - p = 0.6$; r is the requested count; μ, σ^2 are the mean and variance. The stated independence and common p justify $X \sim \text{Bin}(5, 0.4)$.

$$P(X = 2) = \frac{5!}{2!3!} (0.4)^2 (0.6)^3 = 10(0.16)(0.216) = \boxed{0.3456},$$

$$P(X \leq 1) = \binom{5}{0} (0.4)^0 (0.6)^5 + \binom{5}{1} (0.4)(0.6)^4 \\ = 0.07776 + 0.25920 = \boxed{0.33696},$$

$$P(X \geq 1) = 1 - (0.6)^5 = \boxed{0.92224},$$

$$\mu = 5(0.4) = \boxed{2}, \quad \sigma^2 = 5(0.4)(0.6) = \boxed{1.2}.$$



Graph marks: For “at most one,” highlight bars at 0 and 1. For “at least one,” highlight bars at 1 through 5.

Worked question 6: Recover n and p from mean and variance

Question. A binomial random variable has mean 4 and variance 3. Find its parameters.

Formula first. $\mu = np$, $\sigma^2 = np(1-p)$, hence $q = \sigma^2/\mu$, $p = 1 - q$ and $n = \mu/p$ for $\mu > 0, p > 0$. **Symbols.** $\mu = 4$ is the mean; $\sigma^2 = 3$ the variance; n the trial count; p the success probability; $q = 1 - p$ the failure probability.

$$q = \frac{3}{4}, \quad p = 1 - \frac{3}{4} = \frac{1}{4}, \quad n = \frac{4}{1/4} = \boxed{16}.$$

Check: $16(1/4) = 4$ and $16(1/4)(3/4) = 3$. A recovered n must be an integer and p must lie in $[0, 1]$; otherwise the stated exact moments cannot describe this model.

Worked question 7: Smallest number of trials for a target probability

Question. Independent attempts succeed with probability 0.2. Find the smallest number of attempts giving at least a 0.95 probability of one or more successes.

Formula first. $P(X \geq 1) = 1 - q^n$; solve $q^n \leq 1 - c$, so $n \geq \frac{\ln(1-c)}{\ln q}$ for $0 < q < 1$. **Symbols.** n is the trial count, $p = 0.2$ the success probability, $q = 0.8$ the failure probability, $c = 0.95$ the target probability, and $\ln$ the natural logarithm.

$$0.8^n \leq 0.05 \Rightarrow n \geq \frac{\ln 0.05}{\ln 0.8} = 13.4251 \dots \Rightarrow \boxed{n_{\min} = 14}.$$

The inequality reverses when dividing by $\ln 0.8 < 0$. Verification: $1 - 0.8^{13} = 0.9450 < 0.95$, while $1 - 0.8^{14} = 0.9560 \geq 0.95$.

7 Poisson distribution: count events in an interval

Write $X \sim \text{Pois}(\lambda)$ when X counts events in a specified interval of time, space, length or area, with expected count $\lambda > 0$.

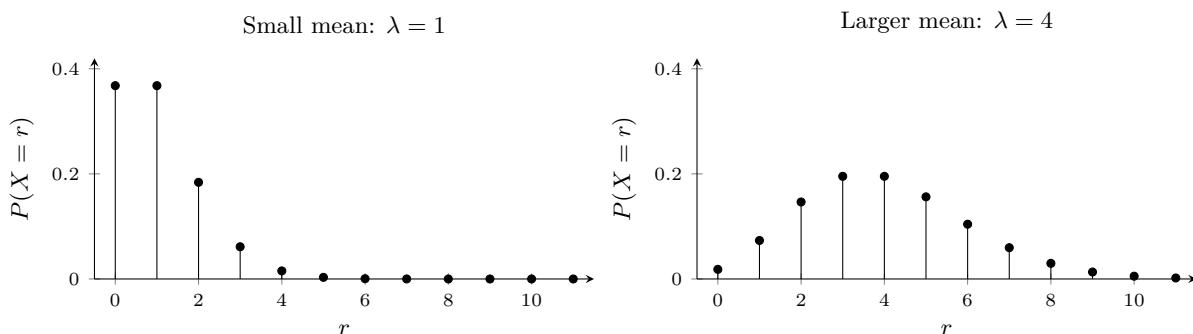
$$P(X = r) = \frac{e^{-\lambda} \lambda^r}{r!}, \quad r = 0, 1, 2, \dots; \quad \mu = \lambda, \quad \sigma^2 = \lambda, \quad \sigma = \sqrt{\lambda}.$$

Here r is a nonnegative integer count; λ is the expected count *in the interval used*; $e \approx 2.71828$ is the exponential constant; $r!$ is factorial with $0! = 1$; μ, σ^2, σ are mean, variance and SD. The special case $\lambda = 0$ puts probability 1 at zero.

Typical modelling conditions: events occur independently; the rate is constant; counts in disjoint intervals are independent; in a sufficiently short interval, two or more events have negligible probability compared with one event. Events arriving in clusters or changing rates may violate the simple model.

$$\lambda = \rho L$$

Here ρ is the rate (events per unit time/length/area) and L is the interval size in the *same unit*. For example, a rate per hour requires L in hours. A Poisson count has no fixed finite upper limit.



Graph marks: Draw discrete masses from zero, with a right tail. As λ grows the shape becomes more nearly symmetric. The plotted range is truncated; the support still continues indefinitely.

Worked question 8: Exactly two, at most one and at least two events

Question. A desk receives a Poisson mean of 3 calls per minute. For one minute, find $P(X = 2)$, $P(X \leq 1)$ and $P(X \geq 2)$, and state the mean and variance.

Formula first. $P(X = r) = e^{-\lambda} \lambda^r / r!$; $P(X \leq 1) = e^{-\lambda}(1 + \lambda)$; $P(X \geq 2) = 1 - P(X \leq 1)$; $\mu = \sigma^2 = \lambda$. **Symbols.** X is calls in one minute; $\lambda = 3$ is the expected count in that minute; r is the requested count; e is the exponential constant; μ, σ^2 are mean and variance.

$$P(X = 2) = e^{-3} \frac{3^2}{2!} = 0.22404,$$

$$P(X \leq 1) = e^{-3} \left(\frac{3^0}{0!} + \frac{3^1}{1!} \right) = 4e^{-3} = 0.19915,$$

$$P(X \geq 2) = 1 - 4e^{-3} = 0.80085, \quad \mu = 3, \quad \sigma^2 = 3.$$

A useful recurrence for making a probability table

$$P(X = 0) = e^{-\lambda}, \quad P(X = r + 1) = \frac{\lambda}{r + 1} P(X = r), \quad r = 0, 1, 2, \dots$$

Here λ is the same interval mean and r the previous count. This follows by dividing consecutive PMF terms; it avoids recomputing every factorial.

Distinguish the constants: binomial p is a probability in $[0, 1]$; Poisson λ is an expected count and can exceed 1. Both distributions describe discrete counts.

8 Poisson interval changes and binomial approximation

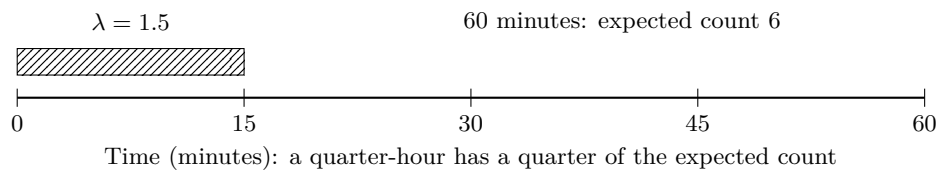
Worked question 9: Change the interval before using the formula

Question. Arrivals follow a Poisson model at 6 per hour. Find the probability of no arrivals and of at least two arrivals during 15 minutes.

Formula first. $\lambda = \rho L$; $P(X = r) = e^{-\lambda} \lambda^r / r!$; $P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)] = 1 - e^{-\lambda}(1 + \lambda)$. **Symbols.** X counts arrivals in 15 minutes; $\rho = 6$ per hour is the rate; $L = 15/60 = 1/4$ hour is the duration; λ is the interval mean; r is an arrival count; e is the exponential constant.

$$\lambda = 6(1/4) = 1.5, \quad P(X = 0) = e^{-1.5} \frac{1.5^0}{0!} = \boxed{0.22313},$$

$$P(X \geq 2) = 1 - e^{-1.5}(1 + 1.5) = \boxed{0.44217}.$$



Binomial versus Poisson: select the model

	Binomial	Poisson
What is fixed?	Number of trials n	Interval and expected count λ
Parameters	n , success probability p	λ
Possible counts	$0, 1, \dots, n$	$0, 1, 2, \dots$
Mean / variance	np / $np(1 - p)$	λ / λ

When n is large and p small, $\text{Bin}(n, p)$ can be approximated by $\text{Pois}(np)$. The limiting statement is $n \rightarrow \infty$, $p \rightarrow 0$, $np \rightarrow \lambda$. There is no universal sharp cutoff for “large” and “small”; use the approximation when requested or clearly justified. It is not an exact equality.

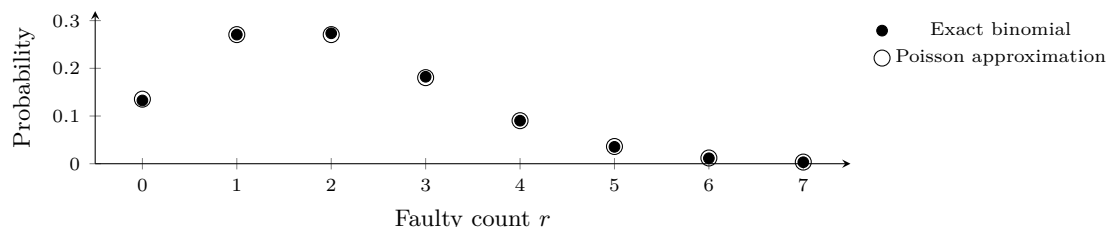
Worked question 10: A rare-event approximation, checked against the exact model

Question. Each of 100 independent components is faulty with probability 0.02. Approximate the probability of exactly two faulty components using Poisson.

Formula first. $\lambda = np$, $P(X = r) \approx e^{-np} (np)^r / r!$; exactly, $P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}$. **Symbols.** X counts faulty components; $n = 100$; $p = 0.02$ is fault probability; $r = 2$ is the requested count; $\lambda = np$ is the approximating Poisson mean.

$$\lambda = 100(0.02) = 2, \quad P(X = 2) \approx e^{-2} \frac{2^2}{2!} = \boxed{0.27067}.$$

Exact check: $\binom{100}{2} (0.02)^2 (0.98)^{98} = 0.27341$; absolute difference ≈ 0.00274 .



9 One-sample t -test: theory and decision rules

Purpose: test a claim about one population mean when the population SD σ is unknown and is estimated by the sample SD s . This compares one sample with a claimed mean, not two samples.

Assumptions: observations form a random, independent sample from the population; the population is normal for the exact small-sample t -test. For a small sample, severe skewness or outliers undermine the procedure. With larger samples it is often approximately usable under suitable conditions; “ $n < 30$ ” is not the definition of a t -test.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}, \quad s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} = \frac{\sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2/n}{n-1},$$

$$s = \sqrt{s^2}, \quad \text{SE} = \frac{s}{\sqrt{n}}, \quad t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad \nu = n - 1.$$

Symbols. x_i is observation i , $i = 1, \dots, n$; $n > 1$ is sample size; $\bar{x}$ is the sample mean; s^2 the sample variance; $s > 0$ the sample SD; SE the estimated standard error of the mean; μ_0 the claimed population mean; t_{obs} the calculated test statistic; and ν (“nu”) the degrees of freedom. Under the null hypothesis and normal-sample assumptions, the random test statistic T has Student’s t distribution with ν degrees of freedom.

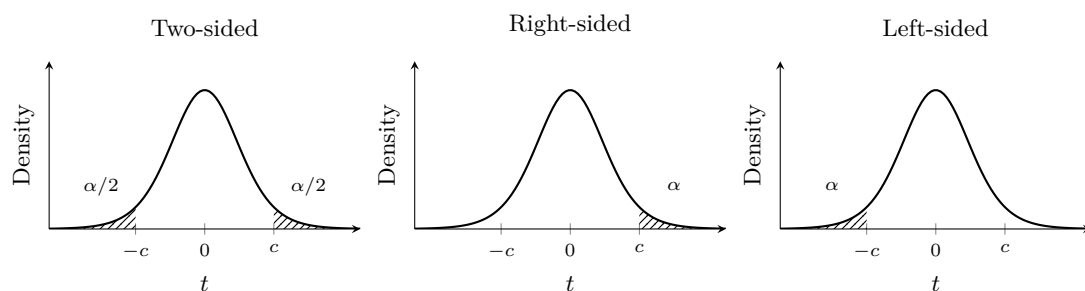
Why $n - 1$? Estimating $\bar{x}$ makes the deviations sum to zero, leaving $n - 1$ freely varying deviations. **Why divide by SE?** The statistic measures how many estimated standard errors separate the observed and claimed means. The t density is symmetric about zero and has heavier tails than the standard normal density.

State the hypotheses and choose the tail before calculating

H_0 is the null hypothesis (test at $\mu = \mu_0$); H_1 is the alternative. Let α be the significance level, usually 0.05. Define $t_{q,\nu}$ by $P(T \leq t_{q,\nu}) = q$; q is a left-tail cumulative probability.

Claim to investigate	H_1	Reject H_0 if	p -value
Different	$\mu \neq \mu_0$	$ t_{\text{obs}} > t_{1-\alpha/2,\nu}$	$2P(T \geq t_{\text{obs}})$
Greater	$\mu > \mu_0$	$t_{\text{obs}} > t_{1-\alpha,\nu}$	$P(T \geq t_{\text{obs}})$
Less	$\mu < \mu_0$	$t_{\text{obs}} < -t_{1-\alpha,\nu}$	$P(T \leq t_{\text{obs}})$

For one-sided tests, the full null can be $\mu \leq \mu_0$ or $\mu \geq \mu_0$; calculations use the boundary μ_0 . The p -value is a tail probability under H_0 , not the probability that H_0 is true. Reject if $p \leq \alpha$.



Graph marks: These are schematic cutoff diagrams: $c = t_{1-\alpha/2,\nu}$ for two sides and $c = t_{1-\alpha,\nu}$ for one side. In your answer, replace c by the correct numerical critical value and mark t_{obs} .

Decision language: “Reject H_0 : sufficient evidence for H_1 ” or “Fail to reject H_0 : insufficient evidence for H_1 .” Failing to reject does not prove the null hypothesis.

10 One-sample t -test: a complete raw-data solution

Worked question 11: Two-sided test at the 5% level

Question. A machine is claimed to dispense a mean of 49 mL. An independent random sample from a normal population gives

48, 49, 50, 51, 52, 50, 49, 51, 50, 50 mL.

The population SD is unknown. Test whether the mean differs from 49 mL at the 5% significance level.

1. Hypotheses. $H_0 : \mu = 49$ mL; $H_1 : \mu \neq 49$ mL. Here μ is the true population mean. “Differs” requires a two-sided test; $\alpha = 0.05$.

2. Formula first, with every symbol.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}, \quad s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}, \quad s = \sqrt{s^2}, \quad t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad \nu = n - 1.$$

Here x_i is the i th volume in mL, $n = 10$ is sample size, $\bar{x}$ is sample mean, s^2 is sample variance in mL², s is sample SD in mL, $\mu_0 = 49$ mL is the claimed mean, $s/\sqrt{n}$ is the standard error in mL, and ν the degrees of freedom. The test statistic is dimensionless.

3. Substitute and calculate.

$$\sum x_i = 500, \quad \bar{x} = 500/10 = 50, \quad \sum (x_i - 50)^2 = 4 + 1 + 0 + 1 + 4 + 0 + 1 + 1 + 0 + 0 = 12.$$

$$s^2 = \frac{12}{9} = \frac{4}{3}, \quad s = \sqrt{\frac{4}{3}} = 1.15470, \quad \text{SE} = \frac{1.15470}{\sqrt{10}} = 0.36515,$$

$$t_{\text{obs}} = \frac{50 - 49}{0.36515} = 2.73861, \quad \nu = 10 - 1 = 9.$$

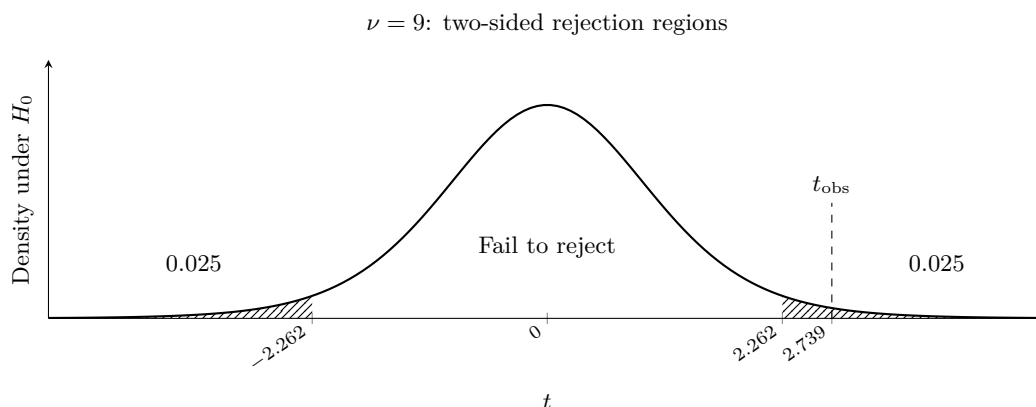
4. Critical values and decision. The rejection formula is

$$|t_{\text{obs}}| > t_{1-\alpha/2, \nu}; \quad t_{0.975, 9} = 2.26216.$$

Here $t_{0.975, 9}$ is the 97.5th percentile of the t distribution with 9 degrees of freedom. Since $2.73861 > 2.26216$, **reject** H_0 .

The equivalent p -value formula is $p = 2P(T_9 \geq |t_{\text{obs}}|)$, where T_9 has a t distribution with 9 degrees of freedom. Numerically $p \approx 0.0229 < 0.05$.

5. Conclusion in context. At the 5% level, there is sufficient evidence that the population mean dispensed volume differs from 49 mL.



Graph marks: Mark both critical values, shade both tails with $\alpha/2 = 0.025$, and place the observed statistic in the right rejection region. The full shaded tails extend to infinity.

Table-reading trap: this two-sided 5% test uses upper-tail probability 0.025, not 0.05. A table labelled “one-tail” uses column 0.025; a table labelled “two-tail” uses column 0.05.

11 One-sided t -tests and final exam checks

Worked question 12: Right-sided test: insufficient evidence

Question. A normal population has unknown SD. An independent random sample of 16 observations gives mean 102 and sample SD 8. At $\alpha = 0.05$, is there evidence that the population mean exceeds 100?

Formula first. $t_{\text{obs}} = (\bar{x} - \mu_0)/(s/\sqrt{n})$; $\nu = n - 1$; reject for $t_{\text{obs}} > t_{1-\alpha, \nu}$; right-tail $p = P(T_\nu \geq t_{\text{obs}})$. **Symbols.** $\bar{x} = 102$ is sample mean, $\mu_0 = 100$ the null mean, $s = 8$ sample SD, $n = 16$ sample size, ν degrees of freedom, $\alpha = 0.05$ significance level; T_ν denotes a t variable with ν degrees of freedom, and $t_{q, \nu}$ its left-tail q quantile.

$$H_0 : \mu = 100, \quad H_1 : \mu > 100; \quad t_{\text{obs}} = \frac{102 - 100}{8/\sqrt{16}} = \boxed{1}, \quad \nu = 15.$$

Critical value $t_{0.95, 15} = 1.75305$. Since $1 < 1.75305$, **fail to reject** H_0 ; $p \approx 0.1666$. There is insufficient evidence at the 5% level that the population mean exceeds 100. A sample mean above 100 is not, by itself, enough evidence.

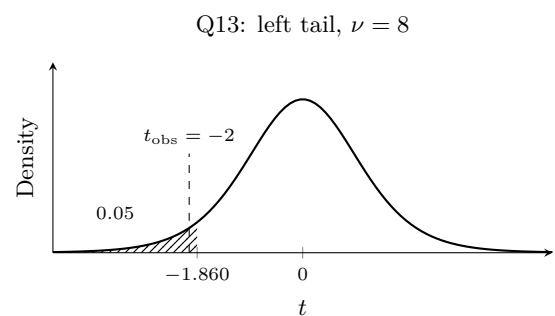
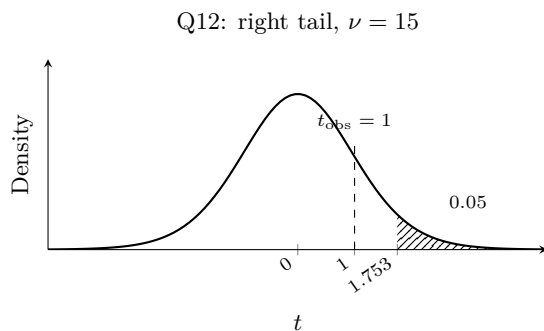
Worked question 13: Left-sided test: evidence of a decrease

Question. Nine independent observations from a normal population give mean 48 and sample SD 3. The population SD is unknown. Test whether the population mean is below 50 at the 5% level.

Formula first. $t_{\text{obs}} = (\bar{x} - \mu_0)/(s/\sqrt{n})$; $\nu = n - 1$; reject for $t_{\text{obs}} < -t_{1-\alpha, \nu}$; left-tail $p = P(T_\nu \leq t_{\text{obs}})$. **Symbols.** $\bar{x} = 48$ is sample mean, $\mu_0 = 50$ null mean, $s = 3$ sample SD, $n = 9$ sample size, ν degrees of freedom, $\alpha = 0.05$ significance level; T_ν is a t variable and $t_{q, \nu}$ its left-tail q quantile.

$$H_0 : \mu = 50, \quad H_1 : \mu < 50; \quad t_{\text{obs}} = \frac{48 - 50}{3/\sqrt{9}} = \boxed{-2}, \quad \nu = 8.$$

Critical value $-t_{0.95, 8} = -1.85955$. Since $-2 < -1.85955$, **reject** H_0 ; $p \approx 0.0403$. There is sufficient evidence at the 5% level that the population mean is below 50.



The last two minutes before the exam

- **DRV:** sum probabilities. **CRV:** integrate density; a single point has probability zero. Check normalization and specify support.
- **Expectation:** multiply value by probability/density before summing/integrating. For $E[X^2]$, square x , not the probability.
- **Variance:** subtract the *square of the mean*. Distribution variance uses probability weights; sample variance in a t -test uses denominator $n - 1$.
- **Binomial:** fixed trials, independence, constant p . **Poisson:** adjust the mean to the actual interval before substitution.
- **t -test:** state assumptions and hypotheses, choose the tail, show $s/\sqrt{n}$, use $n - 1$ degrees of freedom, label and shade the graph, then conclude in context.

For every numerical answer: formula + defined symbols + substitution + result + interpretation. Keep extra digits during calculation and round the final probability to about four or five decimal places.